

Structure-Aware RAG: Structured Retrieval Augmented Generation from Noisy Data for Conversational Agents

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Abstract

Large Language Models (LLMs) have been widely adopted in conversational applications. However, their reliance on parametric knowledge limits reliability in real-world scenarios that require dynamic or domain-specific information. Retrieval-Augmented Generation (RAG) addresses this limitation by incorporating external knowledge during generation, but existing text-based and graph-based RAG methods often struggle with noisy or irrelevant contexts. In this work, we propose *Structure-aware Retrieval Augmented Generation (SA-RAG)*, which uses tables as an intermediate structured representation to provide a compact and controllable interface that reduces noise while preserving essential information. We introduce a quality-aware table metadata generation framework that models metadata normalization and effectiveness, improving metadata quality and downstream performance. Furthermore, we explore both training-free and training-based table generation methods. Generation validation and direct preference optimization further improve table quality while maintaining semantic and structural consistency. Experiments on two noisy real-world datasets show that SA-RAG significantly outperforms existing RAG baselines. Our code is publicly available at an anonymous repository ¹.

1 Introduction

Large Language Models (LLMs) have demonstrated strong capabilities and are widely deployed in conversational applications such as customer service, virtual assistants, and question answering systems (Liang and Tong, 2025; Zhang et al., 2020; Adiwardana et al., 2020; Rome et al., 2024). In these settings, LLMs can interpret user intent and generate coherent, context-aware responses, improving user experience while reducing human labor. Beyond simple query–response interactions,

they can support multi-turn dialogues, maintain conversational context, and adapt responses to user preferences or domain needs. This flexibility enables deployment across diverse domains, including healthcare, finance, education, and e-commerce, where they assist with tasks such as recommendation, scheduling, and decision support (Yang et al., 2025; Xie et al., 2026; Reuters, 2025).

Despite their success, relying solely on the parametric knowledge of LLMs is often insufficient in practice. Many real-world scenarios require access to external, dynamic, or domain-specific knowledge that may not be captured during pre-training (Cuconasu et al., 2024; Laban et al., 2024; Bui et al., 2025). For instance, in conversational settings, knowledge-seeking queries are often mixed with irrelevant content (e.g., non-public app messages in Figure 1), requiring models to retrieve relevant facts from noisy external sources rather than rely solely on memorized knowledge. Without such grounding, LLMs frequently generate hallucinated or outdated responses (Hu et al., 2025). Retrieval-Augmented Generation (RAG) addresses this issue by incorporating information retrieved from external knowledge sources during generation (Lewis et al., 2021; Sparck Jones, 1988).

Text-based RAG (Lewis et al., 2021; Sparck Jones, 1988) retrieves relevant passages from unstructured text and appends them to the model input, while graph-based RAG leverages entity–relation structures from semi-structured data to support reasoning (Moreira et al., 2024; Edge et al., 2024; Laban et al., 2024). However, real-world knowledge sources are often noisy and redundant (e.g., greetings or clarifications in conversations, Figure 1), which limits both paradigms. Text-based RAG is sensitive to irrelevant context, often overwhelming the model with noisy evidence. Graph-based RAG (Wang et al., 2025; Edge et al., 2025) depends on accurate

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entity and relation extraction, which is difficult in noisy settings and can lead to incomplete or erroneous graphs. Prior work attempts to convert text into structured records via information extraction, but such methods typically rely on predefined schemas or rules and remain brittle under noisy conversational data, where extraction errors propagate to retrieval (Mausam et al., 2012). These limitations motivate an intermediate representation that can organize information while filtering noise. Compared with free-form text and rigid graphs, tables provide attribute-centric organization and align queries with relevant fields, enabling aggregation of redundant evidence while remaining robust to missing or noisy entries.

Motivated by these existing problems, we propose Structure-aware Retrieval-Augmented Generation (SA-RAG), a new RAG paradigm that uses tables as an intermediate structured representation between unstructured knowledge sources and LLM generation in noisy data. Tables provide an explicit and compact organization of information along meaningful dimensions, enabling effective noise reduction, improved information density, and more controllable grounding for generation. By serving as a structured interface, tables bridge the gap between raw text and fully symbolic representations, offering a favorable trade-off between expressiveness and robustness.

Introducing tables as a core component of RAG raises several unique challenges, which can be formulated as three fundamental questions. (Q1) How can we automatically construct high-quality table metadata that makes the resulting tables usable and interpretable for retrieval and generation? Table metadata critically determines how information is organized, accessed, and aligned with user queries. However, existing approaches either rely on human-crafted metadata, which is labor-intensive and often inconsistent across datasets and evaluations, or on LLM-generated metadata, which frequently suffers from low accuracy and instability. (Q2) How can we reliably normalize information extracted from heterogeneous and noisy conversational data into a consistent metadata schema? In real-world conversations, knowledge sources are fragmented, redundant, and noisy, requiring careful normalization to ensure semantic consistency and robustness. (Q3) In LLM-based table generation, how can we maintain both structural stability and semantic coherence with the source conversations? Since the generated tables directly serve as retrieval resources,

structural failures (e.g., misaligned attributes) or semantics (e.g., hallucinated or misattributed entries) can severely degrade downstream performance.

To address these challenges, we design the SA-RAG framework with the following key components. During table metadata generation, we incorporate quality evaluation mechanisms to assess and update candidate metadata, ensuring that only high-quality and informative structures are retained. In the process, we model normalization quality with an iterative generation process, enabling consistent alignment of heterogeneous information within table cells. During table generation, we provide both a training-free method and a training-based method to ensure the generation quality and semantic consistency at the same time. For the training-free method, we add a semantic and consistency validation to ensure the generation quality. And for the training method, we adopt Direct Preference Optimization (DPO) to guide the model toward high-quality structured outputs with a single training process and ensure the generation results satisfy the above requirements at the same time. Through these designs, SA-RAG significantly improves robustness and generation quality in noisy real-world settings, providing a principled and effective alternative to existing text- and graph-based RAG approaches.

Our contributions can be summarized as follows:

- We propose **Structure-aware RAG (SA-RAG)**, a novel retrieval-augmented generation framework that leverages **tables** as an intermediate structured representation to improve for **conversational agents**.
- We are the first to focus on how to generate a good table in RAG system. We introduce a **quality-aware table metadata generation mechanism** incorporating the role of **normalization quality**. We propose a **table generation strategy** with training-free and training methods, augmented with **semantic and structural consistency constraints**.
- Experimental results on two noisy conversational datasets demonstrate that our approach consistently outperforms strong RAG baselines in terms of **retrieval and generation performance**. And further experiments show our method generates **high-quality tables** compared to other methods.

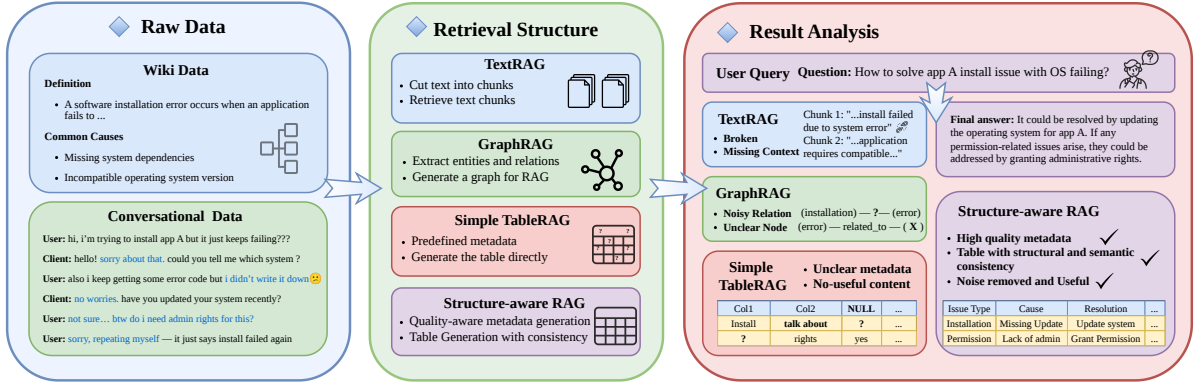


Figure 1: Comparison of SA-RAG with other methods. On clean and well-organized datasets (e.g., Wikipedia-style corpora), most existing RAG methods perform competitively. However, conversational data is inherently noisy, containing irrelevant turns, implicit references, and structural inconsistencies. In such settings, conventional approaches, such as text-based RAG, graph RAG, and simple table RAG, struggle to effectively retrieve and utilize relevant information. In contrast, our Structure-Aware RAG explicitly models and leverages underlying structural signals, enabling robust performance even in the presence of substantial conversational noise.

2 Related Work

2.1 Retrieval-Augmented Generation over Structured Data

Retrieval-Augmented Generation (RAG) typically retrieves unstructured text passages to ground language models. Recent work extends RAG to structured representations such as tables or knowledge graphs, enabling improved evidence aggregation and multi-hop reasoning (Cuconasu et al., 2024; Laban et al., 2024; Bui et al., 2025). Hybrid neural-symbolic approaches further combine neural retrieval with structured operators to improve factual faithfulness and controllability.

Most existing methods focus on leveraging predefined or curated tables during inference, assuming high-quality metadata and static structures (Yin et al., 2020; Herzig et al., 2020). However, they largely overlook how structured representations can be automatically constructed and maintained from noisy, evolving conversational data. In contrast, we propose a metadata quality-aware table construction pipeline that automatically induces, normalizes, and refines relational tables from conversational evidence, enabling reliable table-centric retrieval and generation.

2.2 Content Quality Evaluation in RAG Systems

Data quality plays a critical role in RAG performance, as noisy or redundant evidence can degrade retrieval and lead to hallucinated outputs. Prior work primarily evaluates text-based RAG

using query-aware metrics and improves reliability through reranking or filtering strategies (Shi et al., 2024; Cejas et al., 2025). However, these approaches mainly operate on unstructured documents and do not explicitly evaluate the quality of structured representations.

Our work addresses this limitation by integrating quality evaluation into structure construction. We introduce metadata normalization and metadata effectiveness to assess and regulate metadata quality during table generation, enabling more consistent and effective structured retrieval.

3 Method

3.1 Problem Definition

We consider a conversational retrieval-augmented generation (RAG) setting where the goal is to generate grounded responses to user queries under noisy conversational data.

Input. The input consists of a user query q with a historical conversational data $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$, where N is the number of the conversations. Each conversation d_i is a multi-turn conversation that may contain verbose descriptions, irrelevant context, and redundant information with essential knowledge scattered across turns.

Generation Objective. Given a query q , the system must identify relevant conversations and information to support the response generator in the response generation. The final goal is to generate

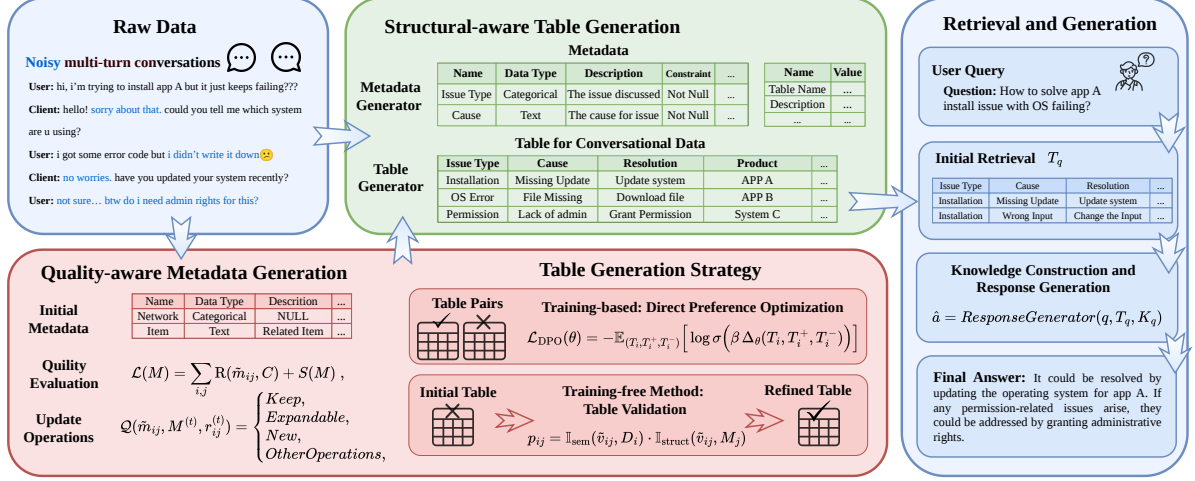


Figure 2: Model Overview. Given a user query and noisy multi-turn conversational history, Structure-Aware RAG first induces latent structured knowledge by generating metadata-aligned tables from raw conversations. The framework then performs joint retrieval over both raw conversations and their structured representations, enabling robust evidence selection under noise and redundancy. Finally, the generator produces a grounded response conditioned on the query and retrieved structured evidence.

a response a that is factually grounded in the retrieved conversational evidence, rather than relying on parametric knowledge and hallucinated content. Formally, we define the task as learning a mapping $f : (q, \mathcal{D}) \mapsto a$, where a is the answer supported by information contained in the historical conversations.

3.2 Model Overview

To address noise and redundancy problems in conversational RAG, we propose *Structure-aware Retrieval Augmented Generation (SA-RAG)*, a unified framework that explicitly models the latent structured knowledge underlying noisy multi-turn conversations with tables. Each conversation $d_i \in \mathcal{D}$ is associated with a structured representation M_i that captures its essential information. Formally,

$$\mathcal{T} = \mathcal{G}_{\text{table}}(\mathcal{D}, M), \quad \mathcal{T} = \{T_1, T_2, \dots, T_N\}.$$

where each T_i is a metadata-aligned set of attribute-value pairs for conversation d_i and M is the metadata. These structures are not directly observable and must be inferred from raw conversational text.

SA-RAG consists of three tightly coupled stages:

1. **Quality-aware Metadata Generation**, which constructs an effective and normalized metadata M^* from noisy conversations;
2. **Table Generation Strategy with Training-Free and Training methods**, which instanti-

ates metadata-aligned tables T_i for each conversation d_i , producing explicit structured representations of latent knowledge;

3. **Retrieval and Generation**, which performs retrieval and generation over conversations and tables. The response generator produces the final response a .

Overall, SA-RAG integrates metadata induction, table generation, and structure-aligned generation into a single framework, explicitly bridging conversational data and grounded response generation.

3.3 Quality-aware Metadata Generation

In the context of a database, metadata is the information that describes the structure, organization, and properties of the data stored within it. Examples of database metadata include table names and column names, data types, relationships between tables like foreign keys, as well as constraints, indexes, views, and stored procedures. High-quality metadata serves as the backbone for reliable table construction and faithful answer generation. Constructing structured table metadata from real-world noisy data is inherently challenging due to noisy language, incomplete description, and highly heterogeneous expressions (Ramakrishnan et al., 2003; Eden, 2002).

Unlike curated databases, noisy data often exhibits inconsistent attribute naming, and varying

levels of granularity, even for semantically identical concepts. As a result, naively aggregating metadata extracted from data tends to produce redundant and low-utility metadata, which severely degrades table generation quality and downstream retrieval-augmented generation performance. However, existing LLM-based extraction approaches typically focus on local correctness and fragmented value, while overlooking global quality and structural compactness. Without explicit quality evaluation, metadata growth becomes uncontrolled, leading to semantic drift, misalignment between the table structure and user queries, and unstructured for the usage.

To address these issues, we explicitly introduce a quality-aware evaluation mechanism that jointly considers semantic coverage, structural coherence, allowing the metadata to evolve in a controlled and interpretable manner. This design enables the metadata to remain both expressive and compact, which is critical for robust table-based retrieval and generation. Given a set of noisy multi-turn conversations $\mathcal{D} = \{d_1, d_2, \dots\}$, our objective is to construct a consistent and high-quality metadata for table generation and downstream retrieval and generation tasks. We explicitly model metadata consistency and metadata evolution through an iterative, quality-aware process.

Initial metadata Extraction. We start from an initial metadata M_0 . For each conversation d_i , an LLM extracts a set of candidate metadata and corresponding information:

$$M_i = \{(m_{ij}, b_{ij}) \mid j = 1, \dots, |M_i|\}, \quad (1)$$

where m_{ij} denotes a candidate column name and b_{ij} its additional information. e.g., type, semantic information. Due to noisy phrasing and heterogeneous expressions, the extracted columns across conversations may be useless, semantically overlapping or inconsistent, which seriously restricts the table quality and the downstream performance.

Metadata Quality Evaluation and Update. To improve metadata consistency and quality, we normalize extracted column names by mapping them to canonical forms following the practice of the database using LLMs. Formally, we use a normalization function $\mathcal{N} : m_{ij} \mapsto \tilde{m}_{ij}$, which resolves synonymy, granularity mismatch, and formatting variations. Applying $\mathcal{N}$ to all extracted columns yields a normalized metadata set for each conversation: $\tilde{M}_i = \{(\tilde{m}_{ij}, b_{ij}) \mid (m_{ij}, b_{ij}) \in C_i\}$. Then,

we formulate metadata induction as an iterative optimization problem that balances coverage and compactness. Let $M^{(t)}$ denote the metadata at iteration t , and $\tilde{m}_{ij}$ be a normalized column extracted from the i -th conversation. We define the following objective function:

$$\mathcal{L}(M) = \sum_{i,j} R(\tilde{m}_{ij}, C) + S(M), \quad (2)$$

where $R(\cdot, \cdot)$ measures how well M could cover the useful information in $\tilde{m}_{ij}$, S measures structural normalization quality as table metadata.

The optimization proceeds iteratively over $\mathcal{D}$. At each step, a candidate normalized column is admitted only after passing quality validation, which ensures the update does not degrade the objective. $\Delta(\tilde{m}_{ij} \mid M^{(t)}) = \mathcal{L}(M^{(t)} \oplus \{\tilde{m}_{ij}\}) - \mathcal{L}(M^{(t)})$, where $\oplus$ is one of the metadata operations, which are defined later. At iteration t , for each $\tilde{m}_{ij}$, we compute its quality evaluation, e.g., answerability, relevance, and alignment with the current metadata: $r_{ij}^{(t)} = r(M^{(t-1)}, \tilde{m}_{ij})$. The benefit of incorporating $\tilde{m}_{ij}$ into the metadata is measured by its gain. According to the quality of new metadata and the overall structure, we classify each column into one of the following metadata operations.

$$Q(\tilde{m}_{ij}, M^{(t)}, r_{ij}^{(t)}) = \begin{cases} \text{KEEP}, \\ \text{EXPANDABLE}, \\ \text{NEW}, \\ \text{OTHER OPERATIONS}, \end{cases} \quad (3)$$

where Q is the metadata operations to update the metadata, we refer readers to the Appendix A for details and examples of the operation. The metadata is updated by greedily incorporating columns that yield positive improvements: $M^{(t+1)} = M^{(t)} \oplus \{\tilde{m}_{ij} \mid \Delta(\tilde{m}_{ij} \mid M^{(t)}) > 0\}$ to maintain the structure and quality of the metadata. To balance reserving most information and metadata complexity, we set a maximum capacity for the metadata.

Iterative Construction. Processing all conversations sequentially induces a sequence of metadata updates $\{M^{(0)}, M^{(1)}, \dots\}$, with $M^{(0)} = M_0$. After the processing, we will have a metadata M^* , which represents a high-quality, normalized, and consistent metadata for the conversation input.

3.4 Table Generation Strategy with Semantic and Structural Consistency

Structured tables provide a machine-interpretable representation bridging unstructured dialogues and downstream retrieval and generation. High-quality table generation requires both semantic faithfulness and structural consistency with metadata. Errors arise from semantic inconsistencies (hallucinated or weakly supported values) and structural violations (type or schema misalignment), which degrade retrieval and generation (Liu et al., 2025b; Ji et al., 2025). Our table generation strategy addresses these issues via semantic validation, structural constraints, and preference-based optimization.

Given the finalized high-quality metadata $M^* = \{m_1, m_2, \dots, m_{|M|}\}$, our goal is to generate a structured table for each conversation D_i that is both semantically faithful to the original conversation and structurally consistent with the metadata. Each table row T_i corresponds to a single conversation and is represented as a set of attribute–value pairs:

$$T_i = \{(m_j, v_{ij}) \mid m_j \in M^*\}. \quad (4)$$

Considering the computational cost, we propose both training-free and training-based generation strategies to improve the semantic and structural consistency.

Training-free Table Generation. In the training-free setting, table values are generated directly by an LLM. For each conversation D_i and each metadata column $M_j \in M^*$, the value generation process is defined as $\tilde{v}_{ij} = \mathcal{G}_{\text{table}}(d_i, c_j)$, yielding a candidate table $\tilde{M}_i = \{(m_j, \tilde{v}_{ij})\}$.

To ensure table quality, we apply *semantic* and *structural* validation. A generated value $\tilde{v}_{ij}$ is retained only if it is supported by the conversation D_i and conforms to metadata m_j :

$$p_{ij} = \mathbb{I}_{\text{sem}}(\tilde{v}_{ij}, D_i) \cdot \mathbb{I}_{\text{struct}}(\tilde{v}_{ij}, m_j), \quad v_{ij} = p_{ij} \odot \tilde{v}_{ij}.$$

Values failing either check ($p_{ij} = 0$) are treated as missing (NULL). The resulting validated table is T_i .

Training-based Table Generation. While training-free generation enforces constraints post-hoc, it cannot proactively shape outputs, leading to low-quality or borderline tables in ambiguous cases. To address this, we adopt a training-based

approach using Direct Preference Optimization (DPO) (Rafailov et al., 2024), which aligns table generation with both semantic faithfulness and metadata consistency. For each conversation d_i , we construct a preference pair (M_i^+, M_i^-) , where M_i^+ is a high-quality human-annotated table and M_i^- is generated via controlled perturbations introducing semantic or structural noise. DPO contrasts these pairs to provide a fine-grained learning signal, enabling globally coherent and robust table generation (details in Appendix A).

Given a policy model π_θ and a frozen reference model π_{ref} , the DPO objective is defined as

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(T_i, T_i^+, T_i^-)} \left[\log \sigma(\beta \Delta_\theta(T_i, T_i^+, T_i^-)) \right], \quad (5)$$

$$\Delta_\theta(T_i, T_i^+, T_i^-) = \log \frac{\pi_\theta(T_i^+ | T_i, M^*)}{\pi_{\text{ref}}(T_i^+ | T_i, M^*)} - \log \frac{\pi_\theta(T_i^- | T_i, M^*)}{\pi_{\text{ref}}(T_i^- | T_i, M^*)}. \quad (6)$$

where β is a temperature hyperparameter.

Optimizing this objective encourages the model to generate tables that are preferred in terms of both semantic fidelity to the conversation and structural consistency with M^* . After training, the model is used to generate the final table collection $\mathcal{T} = \{T_1, T_2, \dots\}$, where each T_i is a high-quality structured row of the table corresponding to conversation D_i .

3.5 Structure-aware Retrieval-Augmented Generation

Given the structured tables $\mathcal{T}$ and a query q , we perform dense retrieval by embedding tables and query into a shared space and selecting top candidates. Information extraction proceeds iteratively: at step t , active values $E^{(t)}$ and relations $R^{(t)}$ are expanded via table lookups, forming the accumulated information $I^{(t)}$. The process terminates when sufficient evidence is collected, yielding structured triples $K_q = \{(e, r, i)\}$ grounded in table entries. The final response is generated by conditioning the LLM on q and K_q , ensuring outputs are grounded in retrieved and verified structured evidence:

$$\hat{a} = \text{Response Generator}(q, \mathcal{T}_q, K_q).$$

4 Experiments

4.1 Experiment Setup

We describe the experimental setup below; additional details are provided in Appendix A and B.

Model	MSDialog						CSDialog					
	Recall	↑%	MRR	↑%	Correctness	↑%	Recall	↑%	MRR	↑%	Correctness	↑%
TextRAG-freq	0.52	—	0.40	—	0.51	—	0.23	—	0.14	—	0.22	—
TextRAG-emb	0.70	34.6%	0.50	25.0%	0.70	37.3%	0.14	-39.1%	0.08	-42.9%	0.14	-36.4%
GraphRAG-KG	0.18	-65.4%	0.15	-62.5%	0.17	-66.7%	0.04	-82.6%	0.02	-85.7%	0.02	-90.9%
GraphRAG-rel	0.14	-73.1%	0.09	-77.5%	0.13	-74.5%	0.02	-91.3%	0.01	-92.9%	0.02	-90.9%
IE-rel	0.32	-38.5%	0.22	-45.0%	0.30	-41.2%	0.24	4.3%	0.16	14.3%	0.22	0.0%
IE-rule	0.12	-76.9%	0.09	-77.5%	0.12	-76.5%	0.32	39.1%	0.19	35.7%	0.29	31.8%
TableRAG-pre	0.18	-65.4%	0.15	-62.5%	0.16	-68.6%	0.26	13.0%	0.18	28.6%	0.25	13.6%
TableRAG-dir	0.42	-19.2%	0.27	-32.5%	0.41	-19.6%	0.31	34.8%	0.25	78.6%	0.31	36.4%
SA-RAG-qwen	0.72	38.5%	0.67	67.5%	0.70	37.3%	0.41	78.3%	0.28	100.0%	0.40	81.8%
SA-RAG	0.94	80.8%	0.82	105.0%	0.92	80.4%	0.53	130.4%	0.42	200.0%	0.52	136.4%

Table 1: Comparison of different models on two datasets across three metrics with improvements (relative to Text-based RAG). We use Recall, MRR, and Correctness for evaluation, and all three metrics are the higher, the better.

Application Scenario and Datasets We study information-seeking conversational agents, where users describe problems in natural dialogue and systems must organize and retrieve historical conversations to generate grounded responses. Such scenarios commonly arise in technical support and customer service. We evaluate on two datasets: **MSDialog** and the **Customer Service Dialogue Dataset** (Qu et al., 2018; Yang et al., 2018; Qu et al., 2019). MSDialog is a public corpus of information-seeking conversations, while the Customer Service dataset contains professional customer-service interactions where users report issues and agents provide solutions.

Baselines We compare against representative RAG baselines across four categories. **Text-based RAG** retrieves unstructured documents, including Text-freq (term-frequency retrieval) and TextRAG-emb (dense embedding retrieval) (Lewis et al., 2021; Sparck Jones, 1988). **Graph-based RAG** retrieves knowledge through graph structures, including GraphRAG-KG and GraphRAG-Rel (Wang et al., 2025; Edge et al., 2025). **Information extraction (IE)** methods construct structured knowledge from text, including IE-rel and IE-rule (Yates et al., 2007; Stanovsky et al., 2018). **Table-based RAG** operates on tabular representations, including TableRAG-pre and TableRAG-dir (Herzig et al., 2020; Liu et al., 2022).

Implementation and Evaluation We use LLaMA-70B (Llama-3.3-70B-Instruct) and Qwen-1.5B (Qwen2-1.5B) as base models. For LLaMA-70B we apply training-free table generation, while Qwen-1.5B is trained with the proposed table generation objective. GPT-4o is used as an *LLM-as-a-judge* with partial human verification.

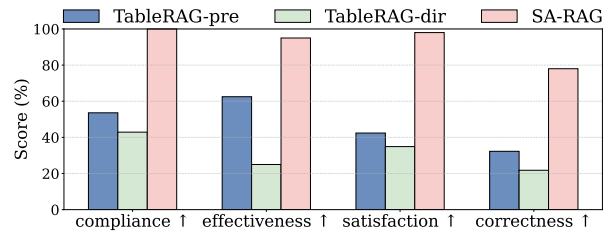


Figure 3: Comparison of Table Quality Metrics Across Different Table Construction Methods. In each subfigure, we present one evaluation result for one metric. For the four metrics, the higher, the better.

We evaluate with **Recall** and **MRR** to measure retrieval effectiveness (Lewis et al., 2021; Karpukhin et al., 2020), and **Result Correctness** to assess answer factuality with respect to the retrieved evidence. The latter follows an LLM-as-a-judge protocol with manual validation (Liu et al., 2025a; Ho et al., 2025).

4.2 Main Results and Analysis

We evaluate our method on the two datasets; results are shown in Table 1. Across MSDialog and CS-Dialogue, **SA-RAG** consistently outperforms all baselines in Recall, MRR, and correctness. The gains stem from the structure-aware retrieval and generation framework rather than model scale. Performance on CSDialogue is lower overall due to longer, more complex interactions, yet SA-RAG achieves notable improvements even in training-free settings. For smaller models, training-based table generation further narrows the gap, benefiting efficiency-constrained scenarios.

Text-based RAG provides limited gains due to noisy or weakly relevant text. SA-RAG mitigates this by converting retrieved evidence into compact tables, filtering irrelevant information while preserving essential attributes. Graph-based RAG suf-

fers from incomplete schemas and error propagation, especially in conversational data with implicit relations. Simple table generation underperforms due to hallucinated or incomplete values. In contrast, SA-RAG leverages quality-aware metadata, validation, and preference-based optimization, improving table accuracy, structural consistency, and downstream response quality.

4.3 In-depth Analysis about the Table Quality Results

To better understand the effectiveness of our **SA-RAG** framework, we conduct an in-depth evaluation of the quality of generated tables. We assess table quality using four complementary metrics that capture structural validity, constraint satisfaction, and semantic accuracy: (1) **structural compliance of metadata**, (2) **metadata effectiveness for downstream tasks**, (3) **value-level constraint satisfaction**, and (4) **semantic correctness of table values**. Please refer to Appendix A for the details about how they are defined and calculated. Figures 3 summarize the performance of different methods across all four metrics. Overall, our method consistently outperforms both baseline approaches on every metric. Specifically, we observe clear gains in metadata evaluation, indicating improved adherence to the predefined metadata, and higher metadata effectiveness, suggesting that the generated structure better supports downstream generation. In addition, our approach achieves stronger value-level constraint satisfaction, reducing invalid entries, and improves semantic correctness, reflecting better factual alignment with the dialogue context. The high-quality meta could improve the consistency of table value generation as it provides enough constraints to generate the values.

4.4 Ablation Study

To assess each component’s contribution, we conduct an ablation study with three SA-RAG variants on the same datasets and metrics. **SA-RAG w/o m** removes metadata construction, generating tables without normalization or effectiveness modeling. **SA-RAG w/o v** disables table validation, using all generated tables regardless of quality. **SA-RAG w/o d** removes DPO fine-tuning, relying on supervised or zero-shot generation.

As shown in Table 2, all removals degrade performance. Omitting metadata (w/o m) most strongly reduces Recall and Eval, highlighting its role in structural guidance. Disabling validation (w/o v)

Method	Recall $\uparrow$		MRR $\uparrow$		Eval $\uparrow$	
	Value	$\uparrow\%$	Value	$\uparrow\%$	Value	$\uparrow\%$
SA-RAG (Llama)	0.94	-	0.82	-	0.92	-
SA-RAG (Gemini)	0.92	-2%	0.81	-1%	0.90	-2%
SA-RAG w/o m	0.45	-52%	0.46	-44%	0.42	-54%
SA-RAG w/o v	0.87	-7%	0.72	-12%	0.71	-23%
SA-RAG w/o d	0.21	-78%	0.17	-79%	0.12	-87%

Table 2: Ablation study of SA-RAG. We report performance and relative drop compared to the full SA-RAG (Llama).

lowers quality by allowing semantically inconsistent tables. Removing DPO (w/o d) causes the largest overall drop, especially for smaller models, indicating its importance in aligning table generation with downstream objectives.

4.5 Different LLM Backbone

To evaluate the generality of SA-RAG across different language models, we conduct experiments using an alternative backbone, Gemini (gemini-3-pro-preview). As shown in the Table 2, The performance of SA-RAG remains stable across both backbones. All three key metrics maintain high scores, indicating that the proposed framework is robust to the choice of underlying LLM. These results suggest that the improvements provided by SA-RAG primarily arise from the structure-aware retrieval and quality-aware table generation, rather than the specific characteristics of a particular backbone model.

4.6 Case Study

To illustrate the framework, the model first extracts high-quality metadata from the conversation and generates structured tables aligned with this metadata in an offline process. During online inference, it retrieves relevant table rows and conversation snippets conditioned on the user query, and produces a grounded response. For example, when asking about resolving an installation error, the system identifies the software, operating system, error code, and suggested fixes from the structured table, and generates a factual answer grounded in both the table and the conversation context. A detailed procedure case is provided in Appendix C.

5 Conclusion

We propose *Structure-aware Retrieval-Augmented Generation (SA-RAG)*, which introduces tables as an intermediate structured representation with quality-aware metadata generation. By providing

a compact and normalized interface, SA-RAG effectively reduces noise while preserving essential information.

Limitations

A limitation of our method is the additional cost of inducing structured representations from raw conversational data. Compared with conventional RAG pipelines that operate directly on text, our approach requires a preprocessing stage to generate metadata and tables. This introduces extra computational overhead during data preparation. However, since this process is performed offline and only once per corpus, it does not impact the latency of online inference. We view this as a practical trade-off for improved robustness under noisy conversational data.

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A Implement Details

We use **LLaMA-70B** (Llama-3.3-70b-instruct) and **Qwen-1.5B** (Qwen2-1.5B) as our base models. **GPT-4o** (gpt-4o) is employed as the *LLM-as-a-judge*, with human verification conducted on a subset of examples to ensure evaluation reliability. For LLaMA-70B, we adopt a training-free table generation approach, while Qwen-1.5B uses a training-based method to generate structured tables from dialogues. During online inference, the models retrieve relevant table rows and conversation snippets conditioned on the user query, and produce grounded answers. Experiments are conducted on two datasets covering diverse conversational and retrieval-augmented generation (RAG) scenarios. Performance is evaluated using three metrics that jointly assess retrieval quality and answer generation: **Recall@3**, **Mean Reciprocal Rank@3 (MRR@3)**, and **Result Correctness**.

Recall@3 For a query q with a set of relevant items R_q , and a model returning the top-3 retrieved items $\text{top}_3(q)$, Recall@3 is defined as:

$$\text{Recall@3}(q) = \frac{|\text{top}_3(q) \cap R_q|}{|R_q|}$$

The overall Recall@3 across all queries Q is the average:

$$\text{Recall@3} = \frac{1}{|Q|} \sum_{q \in Q} \text{Recall@3}(q)$$

MRR@3 For a query q with at least one relevant item, let $\text{rank}_q(i)$ denote the rank of item i in the top-3 retrieved list. The reciprocal rank for q is:

$$\text{RR@3}(q) = \frac{1}{\min\{\text{rank}_q(i) \mid i \in \text{top}_3(q) \cap R_q\}}$$

The Mean Reciprocal Rank@3 (MRR@3) is the average over all queries:

$$\text{MRR@3} = \frac{1}{|Q|} \sum_{q \in Q} \text{RR@3}(q)$$

Result Correctness measures the factual accuracy and informativeness of the generated answers with respect to the retrieved evidence and the user query. It is scored using the *LLM-as-a-judge* protocol, with manual verification on a subset of examples to ensure evaluation quality.

Table Quality Evaluation

To evaluate the quality of tables generated by our **SA-RAG** framework, we compute four metrics using concrete procedures:

1. Structural compliance of metadata:

- For each generated metadata, we check that all required information exist and are named correctly according to the structure requirement.
- A metadata receives a score equal to the fraction of columns that are present and correctly generated.

2. Metadata effectiveness for downstream tasks:

- For each column, we attempt to retrieve relevant information given sample user queries from the dialogue.
- If the column allows successful retrieval and provide useful information, the column is considered effective.

3. Value-level constraint satisfaction:

- We define constraints for each column, such as type constraints, value ranges, allowed categories, and format consistency.
- Each table cell is checked against its column constraints.
- The score for this metric is the fraction of cells that satisfy all constraints.

4. Semantic correctness of table values:

- Each table value is compared to the original dialogue to verify factual consistency.
- Incorrect or hallucinated values are flagged.
- The semantic correctness score is calculated as the proportion of table entries that are factually consistent with the dialogue context.

Finally, for each table, we average the four metric scores to obtain an overall table quality score. We perform this evaluation across all generated tables and compare the results of SA-RAG with two baselines: (1) predefined table generation and (2) direct table construction. This allows us to

quantify improvements in structural validity, usability, and semantic fidelity of the generated tables. These metrics collectively assess both the structural and functional quality of the generated tables. Structural compliance ensures that the tables follow the required schema, which is critical for downstream tasks that rely on predictable formats. Metadata effectiveness measures whether the generated columns actually help answer user queries, reflecting the practical utility of the tables. Value-level constraint satisfaction guarantees that individual entries adhere to expected types and ranges, reducing errors and improving reliability. Finally, semantic correctness ensures that table contents are factually consistent with the source dialogue, preventing hallucinations and maintaining trustworthiness. Evaluating across these dimensions provides a comprehensive picture of table quality, highlighting improvements in usability, accuracy, and overall reliability achieved by our **SA-RAG** framework. In this subsection, we will introduce our experiment setup, and please refer to Appendix A and B for more information.

Application Scenario and Dataset Introduction

We focus on information-seeking scenarios for conversational agents where users describe problems in free-form dialogue, and systems are required to organize, retrieve, and reason over historical data to generate accurate responses. Typical applications include technical support, customer service, and helpdesk systems, where conversations naturally involve structured elements such as issue types, causes, constraints, and solutions, but these structures are often implicit, incomplete, or noisy in the raw conversation. We use two datasets, MSDialog and Customer Service Dialogue Dataset (Qu et al., 2018; Yang et al., 2018; Qu et al., 2019). The MSDialog dataset is a labeled public dialog corpus of interactions between information seekers and answer providers. The Customer Service Dialogue dataset is a private dataset with dialogues between customers and the service people in a highly professional domain. Customers raise issues, and service personnel provide detailed solutions.

Baselines We evaluate our method against different competitive baselines spanning text-based, graph-based, IE-based and table-based retrieval-augmented generation (RAG) approaches. Text-based RAG methods retrieve and condition solely on unstructured textual documents. We consider two variants: Text-freq, which retrieves documents

based on term-frequency statistics and feeds the top-ranked texts to the LLM, and TextRAG-emb, which performs dense retrieval using text embeddings (Lewis et al., 2021; Sparck Jones, 1988). Graph-based RAG methods leverage a graph as the retrieval source to support the retrieval process. We include GraphRAG-KG, which retrieves entities directly from a knowledge graph, and GraphRAG-Rel, which explicitly retrieves and reasons over relation paths (Wang et al., 2025; Edge et al., 2025). Information extraction (IE) approaches directly extract structured knowledge from raw input text prior to retrieving useful information. We evaluate IE-rel, which extracts relational information using learned relation extraction models, and IE-rule, which relies on rule-based extraction pipelines to construct structured facts (Yates et al., 2007; Stanovsky et al., 2018). Table-based generation methods operate on structured tabular representations. We consider TableRAG-pre, which retrieves relevant tables from predefined data and treats them as input to the LLM for generation, and TableRAG-dir, which directly generates table conditioned on conversational data (Herzig et al., 2020; Liu et al., 2022).

Online Retrieval

Given the structured table collection $\mathcal{T} = \{T_1, T_2, \dots\}$ and a user query q , we perform retrieval and reasoning with the high-quality metadata and structured table, explicitly modeling the knowledge over structured tables and conversations before final generation.

We first perform dense retrieval over table rows. Each table M_i and the query q are embedded into a shared vector space via an encoder $\phi(\cdot)$. The initial candidate set is defined as $\mathcal{T}_q^{(0)} = \text{TopK}_{T_i \in \mathcal{T}} \text{sim}(\phi(q), \phi(T_i))$, where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity. We formalize information extraction as an iterative state transition process. At step t , the information set is $S^{(t)} = (E^{(t)}, R^{(t)}, I^{(t)})$, where $E^{(t)}$ denotes the active values set, $R^{(t)}$ the activated relations, and $I^{(t)}$ the accumulated information. The initial state $S^{(0)}$ is constructed from $\mathcal{M}_q^{(0)}$ and q .

For an value $e \in E^{(t)}$ and relation r , the corresponding information is obtained by $v = \text{Tail}(e, r)$, where $\text{Tail}(\cdot)$ retrieves the table entry. Values are aggregated as $I^{(t+1)} = I^{(t)} \cup \{v\}$. To discover additional evidence, we expand the reasoning state by selecting new values at each step. A

relation scoring function evaluates candidate relations conditioned on the query and the current state. The selected relation is added to the active set, and values connected through this relation are activated, forming the value set for the next iteration, i.e., $E^{(t+1)} = \{e' \mid \exists e \in E^{(t)}, \text{Tail}(e, \text{rel}^{(t)}) = e'\}$. The information search process terminates when either a maximum depth is reached or sufficient evidence has been accumulated. Upon termination, we construct a structured evidence set $K_q = \{(e, r, i) \mid e \in E^{(t)}, r \in R^{(t)}, i \in I^{(t)}\}$, explicitly representing value–relation–information triples grounded in table structures.

The final response is generated by conditioning the LLM on the query and the constructed structured knowledge:

$$\hat{a} = \text{ResponseGenerator}(q, T_q, K_q). \quad (7)$$

This formulation ensures that generation is grounded in iteratively retrieved and verified structured evidence.

Metadata Update Operations. To maintain a compact and effective metadata schema, we define several update operations for each normalized metadata candidate $\tilde{m}_{ij}$: Here, KEEP preserves existing metadata (e.g., *issue_type* and *problem_category*), EXPANDABLE extends existing schema semantics (e.g., *error_code* $\rightarrow$ *installation_error_code*), and NEW introduces previously unseen but useful metadata fields (e.g., *operating_system_version*). Additional governance operations, such as MERGE and DELETE, maintain metadata quality and compactness.

LLM Prompts for Table and Metadata Generation

Below we provide the prompts used in our pipeline for structured table extraction, schema normalization, quality evaluation, and meta-data evolution. Each prompt is presented in a separate box for clarity.

Prompt 1: Extract Columns from Dialogue

You are a data modeling assistant.

TASK: Given a customer support dialogue, infer a relational database table schema to solve the problem. Provide more than ten columns useful to classify and answer future issues, not redundant information.

INSTRUCTIONS:

- Identify the main table that represents the dialogue.
- Generate a table schema using the following JSON structure ONLY:

```
[
  {
    "table_name": string,
    "Columns": [
      {
        "name": string,
        "type": one of ["int",
          "string",
          "float", "boolean", "date",
          "datetime"],
        "semantic": string,
        "constraints": [string],
        "Note": "Important
          information"
      }
    ],
    "functional_dependencies":
      [string],
    "version": int
  }
]
```

- Column names: snake_case, concise, descriptive.
- Semantic: describe real-world meaning, prefer ontology-style naming.
- Constraints: use primary key, foreign key, not null, unique when applicable.
- Output valid JSON only, no explanations.

DIALOGUE: {dialog}

Prompt 2: Normalize Schema

You are a database schema normalization assistant.

TASK: Normalize the given raw database schema in ONE step.

RULES:

- Column names: snake_case, concise, descriptive.
- Data types: normalize to one of ["int", "string", "float", "boolean", "date", "datetime"].
- Semantic: provide clear standardized meaning, prefer ontology-style naming.
- Constraints: preserve valid constraints, normalize naming.
- Structure: preserve table_name, version, functional dependencies; fix spelling issues.
- Output valid JSON only.

INPUT SCHEMA: {raw_schema}

Prompt 3: Quality Check

You are a database schema quality evaluator.
TASK: Evaluate the quality of EACH COLUMN in the new schema based ONLY on its usefulness for answering the given problem.
COLUMN-LEVEL METRICS (0–1):

1. relevance: How directly the column relates to the problem.
2. answerability: Contribution to answering the problem.
3. overall: Holistic judgment for this problem.

Output: For each column, add field `quality_score` with relevance, answerability, overall, and justification. Return valid JSON only.

CURRENT META DATA: {current_meta_data}

NEW SCHEMA TO EVALUATE: {new_schema}

PROBLEM / QUERY: {problem}

Prompt 4: Schema Governance and Merge

You are a schema governance and evolution assistant.

TASK: Given current meta-data, a new schema with column-level quality scores, and problem context, decide how to evolve the meta-data to retain columns useful for identifying or solving the issue. Every column should be useful.

Steps (perform all in one pass):

1. Decide if the new schema is acceptable.
2. Detect semantic overlap with existing schemas.
3. For each overlap, choose one operation: ADD, UPDATE, MERGE, KEEP, DELETE.
4. Produce the FINAL updated meta-data (same format as current meta-data, without quality scores).
5. Ensure no more than 20 columns.

Decision rules:

- Use `column-level quality_score.overall` as primary signal.
- Prefer schemas with higher average column quality, better coverage, and clearer semantics.
- Delete only strictly worse redundant columns.
- Avoid introducing new tables.

Output: valid JSON ONLY, final meta-data list.

CURRENT META DATA: {current_meta_data}

NEW SCHEMA (WITH QUALITY SCORES): {quality_annotated_schema}

PROBLEM CONTEXT: {problem}

Prompt 5: Generate Data Row from Dialogue

You are a data extraction assistant.
TASK: Generate ONE structured data row from the given dialogue.
IMPORTANT RULES:

- Meta-data is READ-ONLY context.
- Do NOT modify, normalize, rename, or reinterpret the schema.
- Use column names EXACTLY as provided.
- If a value cannot be inferred, output null.
- Follow declared data types strictly.
- Return VALID JSON ONLY.
- No explanations, no comments.

META-DATA: {meta_data}

DIALOGUE: {dialogue}

OUTPUT FORMAT:

```
{
  "row": {
    "<column_name>": <value | null>
  }
}
```

Prompt 6: Judge Table Against Metadata

You are a data validation assistant.
TASK: Determine whether the table satisfies the metadata requirements (e.g., all non-null columns are filled).

META-DATA: {meta_data}

TABLE: {dialogue}

OUTPUT FORMAT:

- Satisfy the metadata: yes or no.
- If no, explain which parts do not satisfy the metadata.

B Dataset Introduction

The MSDialog dataset is a labeled dialog corpus of question-answering (QA) interactions between information seekers and answer providers on the Microsoft Community forum (Qu et al., 2018;

Yang et al., 2018; Qu et al., 2019). It comprises over 2,000 multi-turn information-seeking conversations, totaling approximately 10,000 utterances, each annotated with user intent at the utterance level. Annotations were collected via crowdsourcing using Amazon Mechanical Turk. MSDialog provides several versions, including the full collection (MSDialog-Complete) and a labeled subset (MSDialog-Intent). For user intent annotation, they selected a subset of dialogs from MSDialog-Complete based on the following quality criteria: (1) conversations containing 3 to 10 turns, (2) involving 2 to 4 participants, (3) containing at least one community-verified correct answer, and (4) belonging to one of four major Microsoft product categories: Windows, Office, Bing, and Skype. This filtering resulted in approximately 2,400 dialogs for annotation.

We use the Customer Service Dialogue Dataset, a private multi-turn conversational corpus between customers and service personnel in a high-professional domain. Each dialogue records customers describing issues and service agents providing detailed solutions. While the dialogues often contain structured elements such as issue types, causes, constraints, and suggested resolutions, these structures are typically implicit, incomplete, or noisy in the raw conversation. Due to confidentiality requirements, we cannot disclose specific statistics such as the number of dialogues, turns, or users, but the dataset covers a wide range of real-world customer support scenarios, making it suitable for evaluating information retrieval, structured reasoning, and grounded response generation in conversational agents

Preference Pair Construction

For each conversation d_i , we construct a set of preference pairs (M_i^+, M_i^-) to enable preference-based supervision.

Positive Tables M_i^+ The positive table M_i^+ is obtained from human-annotated, high-quality structured representations. These tables are carefully curated to ensure:

- **Semantic correctness:** all table entries accurately reflect the information in the dialogue.
- **Schema consistency:** column names, data types, and structural completeness conform to the predefined table schema.

Negative Tables M_i^- Based on each M_i^+ , we generate a corresponding negative table M_i^- by applying controlled perturbations to the structured fields. The perturbations are designed to produce tables that are syntactically valid but semantically or structurally misaligned with the dialogue. The methods include:

1. **Random field drop:** randomly set some fields to null to simulate missing values.
2. **Dialogue-based hallucination:** replace some field values with randomly sampled words from the dialogue to produce realistic but incorrect entries.
3. **Field swapping:** swap values across columns to create structural misalignment.
4. **Dialogue-inconsistent fill:** fill fields with randomly selected sentences from the dialogue, teaching the model that direct copying of dialogue may be incorrect.

Combination of perturbations For diversity, one or multiple perturbations are randomly applied per table instance. This ensures that negative tables remain well-formed syntactically while violating semantic or structural alignment, which provides effective supervision signals for preference-based training.

Implementation Details The negative table generation process is implemented in Python. For each dialogue row, we randomly select a perturbation mode from ["drop", "hallucinate", "swap", "dialogue", "combo"]. The combo mode applies multiple perturbations sequentially to create more challenging negative examples. After generating all negative tables, we save them to a CSV file (structure_bad.csv) for downstream use.

- `drop_random_values(row, drop_rate):` randomly sets a fraction of fields to null.
- `random_hallucination(row, noise_rate):` replaces selected fields with randomly sampled words from the dialogue.
- `swap_fields(row, swap_pairs):` swaps values between randomly selected pairs of columns.

- `dialogue_inconsistent_fill(row):` fills a random field with a randomly chosen sentence from the dialogue.

This process produces a diverse set of preference pairs (M_i^+, M_i^-) , which are then used for training and evaluating the model’s ability to distinguish high-quality structured tables from noisy alternatives.

C Case Study

Scenario Overview

This case study illustrates how the **SA-RAG** framework processes a multi-turn customer support conversation from offline table construction to online retrieval and response generation. The conversation is anonymized and simplified to focus on the workflow and structured representation rather than user or device details.

Offline Phase: Metadata and Table Generation

In the offline stage, SA-RAG constructs structured metadata from historical conversations. For each conversation:

- **Metadata Extraction:** The system identifies key attributes of the conversation, such as `issue_title`, `issue_description`, `operating_system`, `symptoms`, and `agent_response`.
- **Normalization:** Extracted fields are normalized according to a predefined schema: column names are in `snake_case`, data types are standardized, and semantic meanings are clarified.
- **Quality Control and update:** Each column is evaluated for relevance, answerability, and overall quality with respect to the conversation context. Low-quality or inconsistent entries are flagged or corrected.
- **Final Table Storage:** After normalization and quality validation, the structured tables are stored in a database to be used for online retrieval.

At the end of this phase, each historical conversation is represented as a structured table row containing the most informative fields, ready to support downstream retrieval tasks.

Online Phase: Retrieval and Grounded Response

When a new user query or ongoing conversation occurs:

- **Query Understanding:** The system interprets the user's question in the context of the ongoing conversation.
- **Table Retrieval:** SA-RAG retrieves relevant rows from the offline-generated tables that best match the current query, considering semantic similarity and relevance.
- **Grounded Answer Generation:** Using the retrieved table rows and the conversation context, the model generates responses that are factual and aligned with the structured metadata.
- **Dynamic Updates:** If a new schema is generated or existing metadata needs adjustment, SA-RAG evaluates the new schema using quality scores, decides whether to add, update, merge, or discard columns, and updates the meta-data for future queries.

Example Walkthrough

Consider a conversation where a user reports a login failure:

1. Offline Table Generation:

- `issue_title`: Login failure
- `issue_description`: Unable to log in, receiving an error about settings
- `operating_system`: Mobile OS
- `symptoms`: Login error message appears
- `agent_response`: Device no longer supported
- `issue_resolution_status`: Unresolved
- Other fields such as `resolution_steps` may be empty if not mentioned.

2. Online Retrieval:

- When a new query about a similar login issue arises, SA-RAG retrieves this row from the offline table.
- The system focuses on the most relevant columns (`issue_title`, `issue_description`, `symptoms`) to generate a grounded response.

3. Grounded Response Generation:

- The model produces a response explaining the issue, referencing the relevant table information, ensuring factual consistency with the conversation history.

This workflow demonstrates how SA-RAG bridges unstructured dialogues and structured reasoning, enabling accurate, context-aware, and explainable responses in conversational information-seeking scenarios.